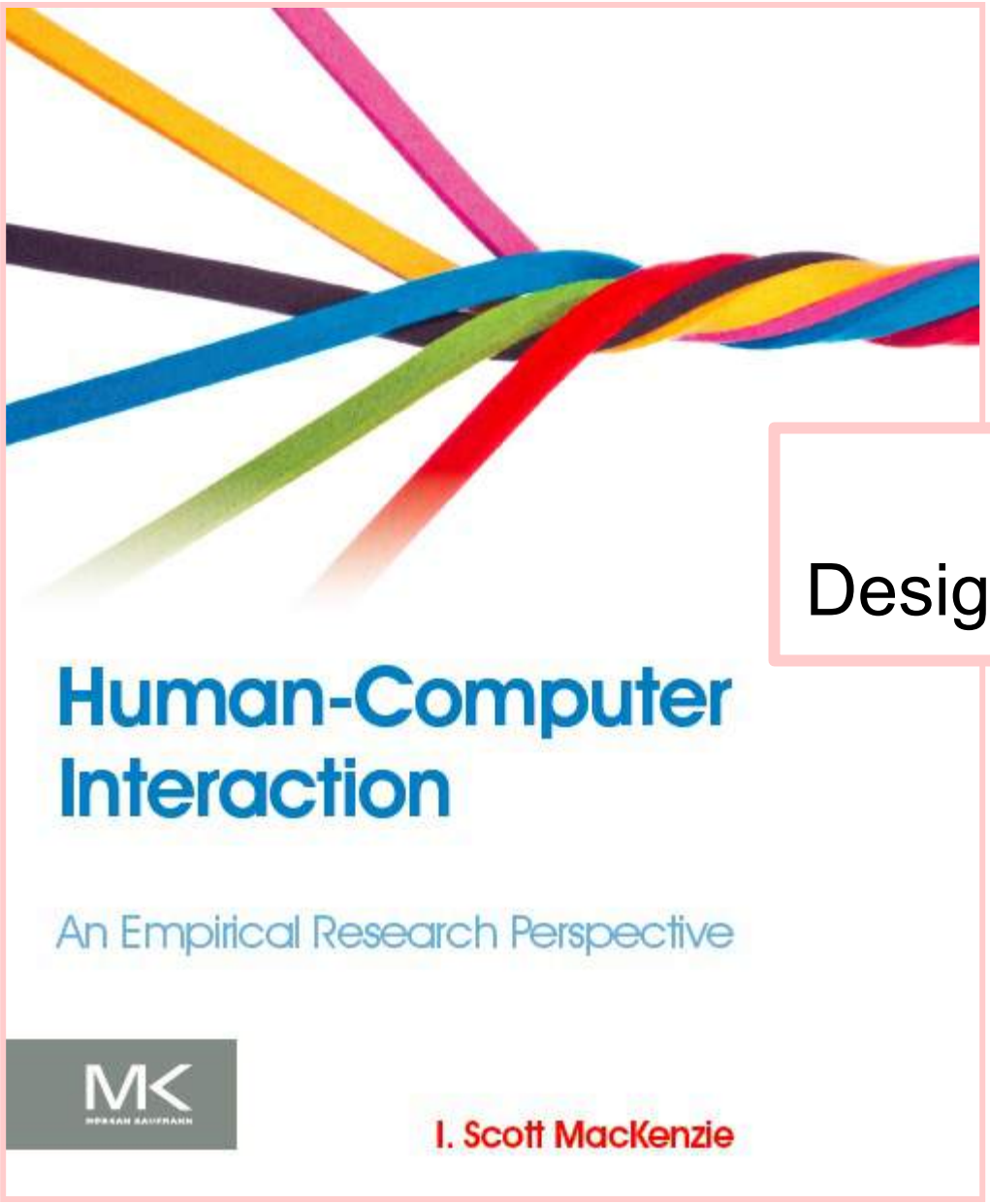
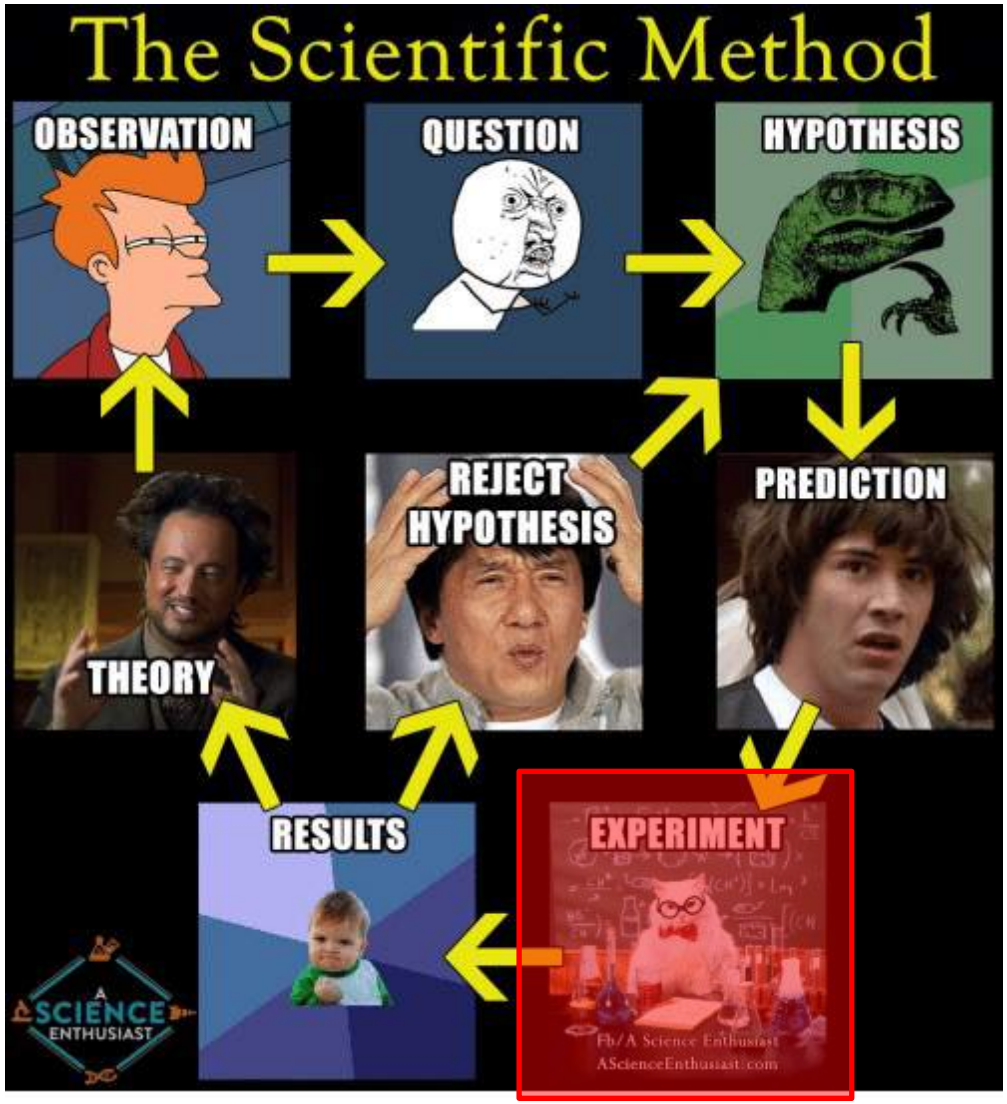


Chapter 5
Designing HCI Experiments

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The Scientific Method



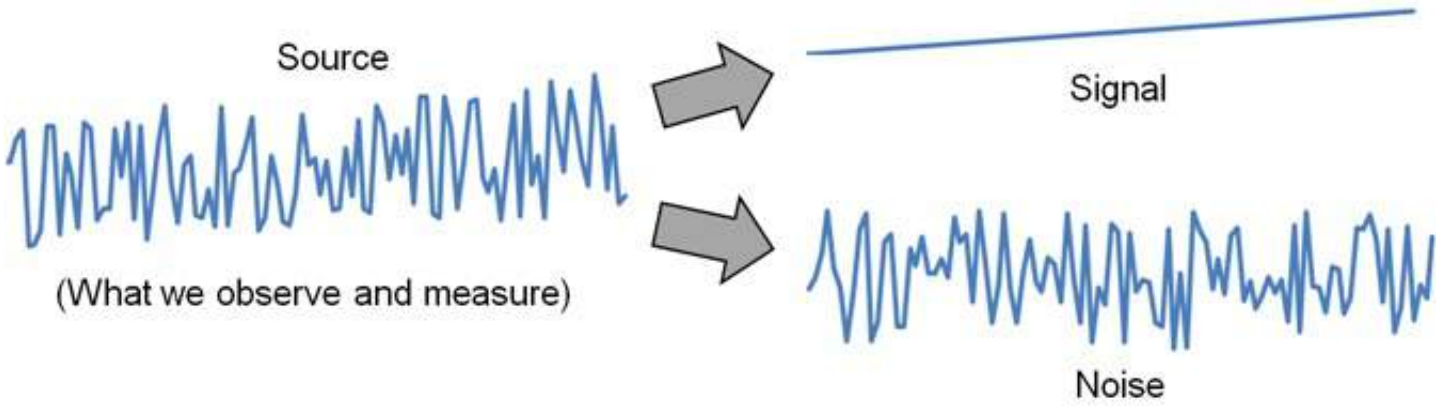
Introduction

- Learning to conduct and design an experiment is a skill required of all researchers in HCI

- ***Experiment design*** is the process of deciding what **variables** to use, what **tasks** and **procedures** to use, how many **participants** to use and how to **solicit** (要求) them, and so on

Signal and Noise Metaphor

- Signal and noise metaphor for experiment design:



- Signal → a variable of interest
- Noise → everything else (random influences)
- Experiment design seeks to **enhance the signal, while minimizing the noise**

Methodology

- *Methodology* is the way an experiment is designed and carried out
- Methodology is critical:

> Science is method. Everything else is commentary.[1]

- What methodology?
- Don't just make it up because it seems reasonable
- Follow standards for experiments with human participants (next slide)

[1] This quote from Allen Newell was cited and elaborated by Stuart Card in an invited talk at the ACM's SIGCHI conference in Austin, Texas (May 10, 2012).

Ethics Approval

- *Ethics approval* is a crucial step that precedes every HCI experiment
- HCI experiments involve humans, thus…

> Researchers must respect the **safety, welfare, and dignity** of human participants in their research and treat them equally and fairly.[1]

- Proposal submitted to **ethics review committee**
- Criteria for approval
  - research methodology
  - risks or benefits
  - the right not to participate, to terminate participation, etc.
  - the right to anonymity and confidentiality

[1] http://www.yorku.ca/research/students/index.html

Getting Started With Experiment Design

- Transitioning from the **creative work** in formulating and prototyping ideas to **experimental research** is a challenge
- Begin with…

What are the experimental variables?

- Remember research questions:

Can a task be performed more quickly with my new interface than with an existing interface?

Getting Started With Experiment Design

- Variables

Independent Variables **(Settings)**	Dependent Variables **(Metrics)**
Control Variables (light, temperature, …)	Random Variables (everything else)

Independent Variable (IV)

- **Definition**: An *independent variable* (IV) is a circumstance or characteristic that is manipulated in an experiment to elicit (引出) a change in a human response while interacting with a computer.
- **Independent**: there is nothing a participant can do to influence an independent variable
- **Examples**:
  - interface, device, feedback mode, button layout, visual layout, age, gender, background noise, expertise, etc.
- synonymous: independent variable (IV), factor

Types of IVs

- **Human characteristics**: naturally occurring attributes
  - Examples: Gender, age, height, weight, handedness, grip strength, finger width, visual acuity, personality trait, political viewpoint, first language, shoe size, etc.

- **Environmental circumstance:**
  - Example: background noise (quiet, noisy), room lighting (sun, incandescent, fluorescent), vibration level (calm, in a car, in a train), etc.

IV: Test Conditions

- An independent variable (IV) must have at least two **levels** (values)
- The levels (values) for an IV are the *test conditions*

IVs	Levels (test conditions)
Device	mouse, trackball, joystick
Feedback mode	audio, tactile, none
Task	pointing, dragging
Visualization	2D, 3D, animated
Search interface	Google, custom

How Many IVs?

- An experiment must have at least one IV (factor)
  - Possible to have 2, 3, or more factors
- The number of "effects" increases rapidly with the number of IVs (size of the experiment):

Independent Variables	Effects					Total
	Main	2-way	3-way	4-way	5-way	
1	1	-	-	-	-	1
2	2	1	-	-	-	3
3	3	3	1	-	-	7
4	4	6	3	1	-	14
5	5	10	6	3	1	25

- Advice: Keep it simple (1 or 2 factors, 3 at the most)

Further details will be discussed in *Hypothesis Testing*

Dependent Variable (DV)

- A *dependent variable* is a **measured human behavior**
- **Dependent**: depends on what the participant does
- Examples:
  - task completion time, speed, accuracy, error rate, throughput, target re-entries, task retries, presses of backspace, etc.

- Dependent variables must be clearly defined
  - *Research must be reproducible!*

Control Variable

- A *control variable* is a circumstance (condition) that is kept **constant** while testing the effect of an independent variable
- More control means the experiment is less **generalizable** (i.e., less applicable to other people and other situations)
- **Example**: *Is there an effect of font color on online reading comprehension?*
  - Independent variables: font color
  - Dependent variable: comprehension test scores
  - Control variables
    - Font size (e.g., 12 point)
    - Font family (e.g., Times)
    - Ambient lighting (e.g., fluorescent, fixed intensity)
    - Etc.

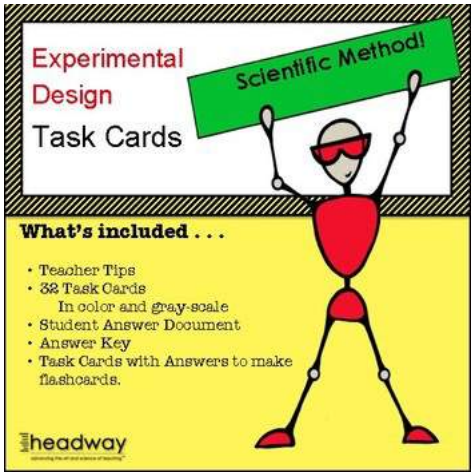
Random Variable

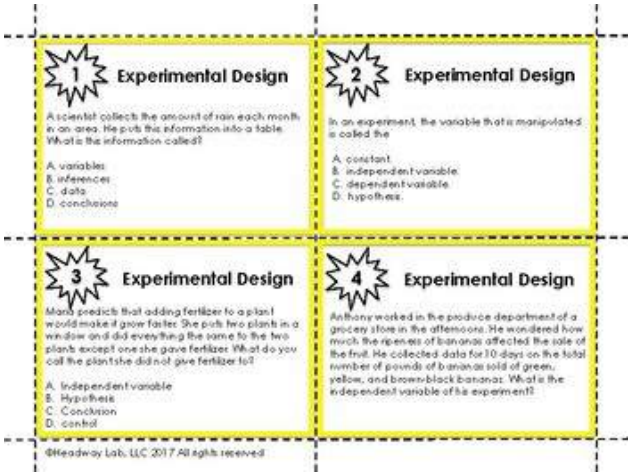
- A *random variable* is a circumstance (condition) that is allowed to vary **randomly**
- More variability is introduced in the measures (that's bad!), but the results are more generalizable (that's good!)
- **Example**: *Does user stance (姿态) affect performance while playing the game Honor of Kings?*
  - Independent variable: stance (standing, sitting)
  - Dependent variable: scores of games
  - Random variables
    - Prior knowledge of the *heroes*
    - Prior experience playing *Honor of Kings*
    - Amount of *coffee* consumed prior to testing
    - …



Experiment Task

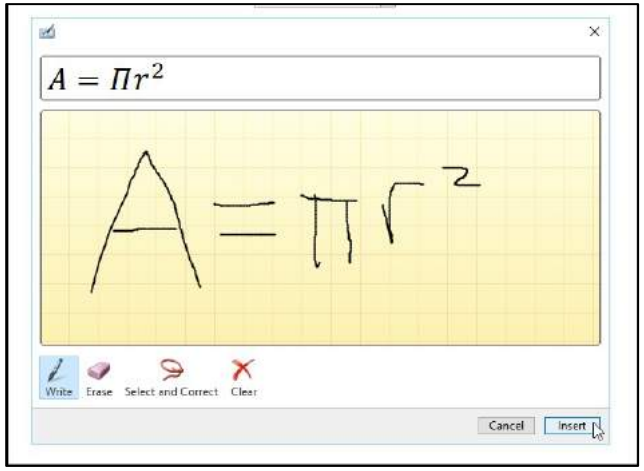
- The experiment task must "**elicit a change**"
  - In a human response while interacting with a computer
- Qualities of a good task: *represent*, *discriminate*
  - **Represent** activities people do with the interface
  - **Discriminate** among the test conditions



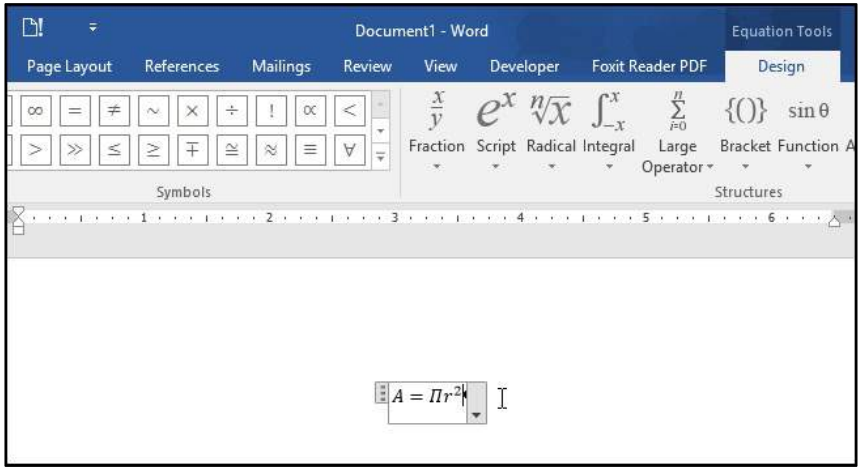


Task Examples

- Usually the task is **self-evident** (follows directly from the research idea)
- **Idea**: *a new graphical method for entering equations in a spreadsheet*
- **Task**: create an equation using



(a) the graphical method

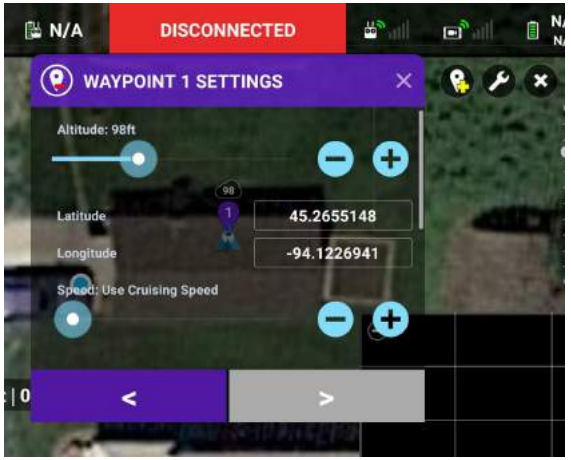


(b) the conventional method

Task Examples

- Usually the task is **self-evident** (follows directly from the research idea)
- **Idea**: *an auditory feedback technique for programming a GPS device*
- **Task**: program a destination location into a GPS device using
  - (a) the auditory feedback method
  - (b) the conventional method



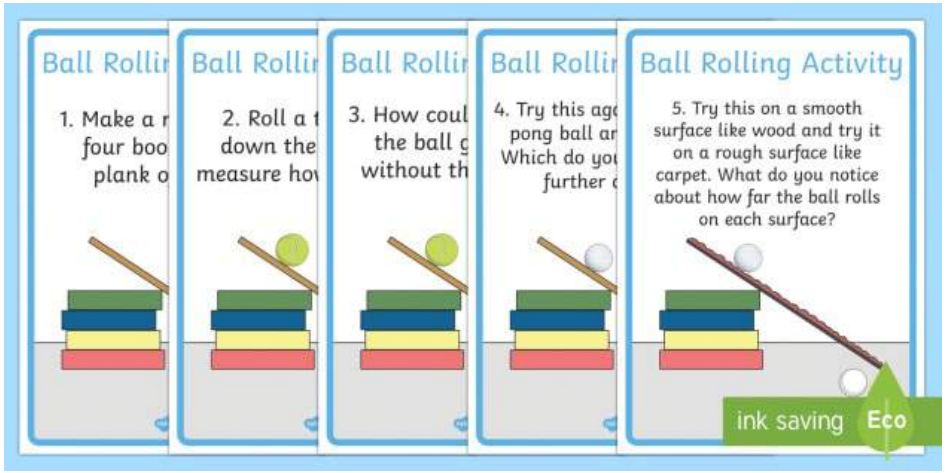


Procedure

- **Procedure:** everything that occurs with participants in the experiment task
  - Arriving, welcoming
  - Signing a consent form (知情同意表)
  - **Instructions** given to participants about the experiment task
  - Demonstration trials, practice trials
  - Rest breaks
  - Management of questionnaires or interviews

Instructions

- Very important (best to prepare in advance; write out)
- Goal: “to proceed as quickly and accurately as possible but at a pace that is comfortable”
- Give the **same** instructions to all participants
- If a participant asks for **clarification**, do not change the instructions in a way that may cause the participant to behave **differently** from the other participants (consistency)



Participants

- Experimental results need to apply to people not actually tested – a **population**
- Examples:
  - Computer-literate adults, teenagers, children, people with certain disabilities, left-handed people, engineers, musicians, etc.
- For results to apply **generally** to a population, the participants used in the experiment must be…
  - **Members** of the desired population
  - Selected at **random** from the population
- True random sampling difficult
  - Some form of **convenience sampling**

How Many Participants?

- Too few → experimental effects fail to achieve statistical significance (显著性)
- Too many → statistical significance for effects, but not practical
- The correct number:
  - Use the same number of participants as used in similar research[1] (empirically)



[1] Martin, D. W. (2004). *Doing psychology experiments* (6th ed.). Pacific Grove, CA. Belmont, CA: Wadsworth.

Questionnaires

- Questionnaires are used in most HCI experiments
- Two purposes
  - **Collect information about the participants**
    - Demographics (gender, age, first language, handedness, visual acuity, etc.)
    - Prior experience related to the current research
  - **Obtain feedback, comments, impressions, suggestions,** etc., about participants' use of the experimental apparatus
- Questionnaires are usually **brief** for clarity

Information Questions

- Questions constructed according to how the information will be used

Please indicate your age:_____

Ratio-scale data

Please indicate your age?

☐ < 20 ☐ 20-29 ☐ 30-39

☐ 40-49 ☐ 50-59 ☐ 60+

Ordinal-scale data

Which browser do you use? _____

Open-ended

Which browser do you use?

☐ Mozilla *Firefox* ☐ Google *Chrome*

☐ Microsoft *IE* ☐ Other (_____)

Closed

Participant Feedback

- Using NASA Task Load Index (TLX):

Frustration: I felt a high level of insecurity, discouragement, irritation, stress, or annoyance.

1	2	3	4	5	6	7
Strongly disagree			Neutral			Strongly agree

- ISO 9241-9:

Eye fatigue:

1	2	3	4	5	6	7
Very high						Very low

Data Collection

- Data for **dependent variables** must be collected
- Ideally, engage the experiment software to log
  - timestamps, key presses, button clicks, etc.
- Planning and pilot testing is important
- Ensure conditions are identified, either in the filenames or in the data columns
- Examples: (next two slides)

TextInputHuffman-P01-D99-B06-S01.sd1

```
my bike has a flat tire
my bike has a flat tire
16 3
891 2
1797 3 m
3656 2
4188 1
4672 2 y
5750 3
5938 3 [Space]
6813 3
6984 2
7219 0
8656 3 b
```

the participant (P01), the device (D99), the block (B06) and the session (S01)

"sd1" for "summary data one."

TextInputHuffman-P01-D99-B06-S01.sd2

```
min_keystrokes,keystrokes,presented_characters,transcribed_characters, ...
55, 59, 23, 23, 29.45, 0, 9.37, 0.0, 2.5652173913043477, 93.22033898305085
61, 65, 26, 26, 30.28, 0, 10.3, 0.0, 2.5, 93.84615384615384
85, 85, 33, 33, 48.59, 0, 8.15, 0.0, 2.5757575757575757, 100.0
67, 71, 28, 28, 33.92, 0, 9.91, 0.0, 2.5357142857142856, 94.36619718309859
61, 70, 24, 24, 39.44, 0, 7.3, 0.0, 2.9166666666666665, 87.14285714285714
```

the participant (P01), the device (D99), the block (B06) and the session (S01)

“sd2” for “summary data two.”

Within-subjects, Between-subjects

- Two ways to assign conditions to participants:
  - **Within-subjects:** each participant is tested on each condition (aka *repeated measures*)
  - **Between-subjects:** each participant is tested on one condition only
  - Example: An IV with three test conditions (A, B, C):

Within-subjects

Participant	Test Condition		
1	A	B	C
2	A	B	C

Between-subjects

Participant	Test Condition
1	A
2	A
3	B
4	B
5	C
6	C

Order Effects, Counterbalancing

- Only relevant for within-subjects factors
- The issue: order effects
  - aka *learning effects*, *practice effects*, *fatigue effects*, *sequence effects*

Participant	Test Condition		
1	A	B	C
2	A	B	C

Participant	Test Condition		
1	B	C	A
2	B	C	A

Order Effects, Counterbalancing

- Order effects offset by **counterbalancing**:
  - Participants divided into groups
  - Different group, different order: one for each
  - Order of test conditions: **Latin square**
    - each condition occurs precisely once in each row and column

A	B	C
B	C	A
C	A	B

Latin Squares

2 x 2

A	B
B	A

3 x 3

A	B	C
B	C	A
C	A	B

4 x 4

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

5 x 5

A	B	C	D	E
B	C	D	E	A
C	D	E	A	B
D	E	A	B	C
E	A	B	C	D

Example

- An experimenter seeks to determine if three *editing methods* (A, B, C) differ in the *amount of time* to do a common editing task:

Replace one 5-letter word with another, **e.g. pizza →crazy**

- 12 participants are recruited and divided into 3 groups (4 participants/group)
- Methods administered using a 3 × 3 Latin Square
- Results (next slide)

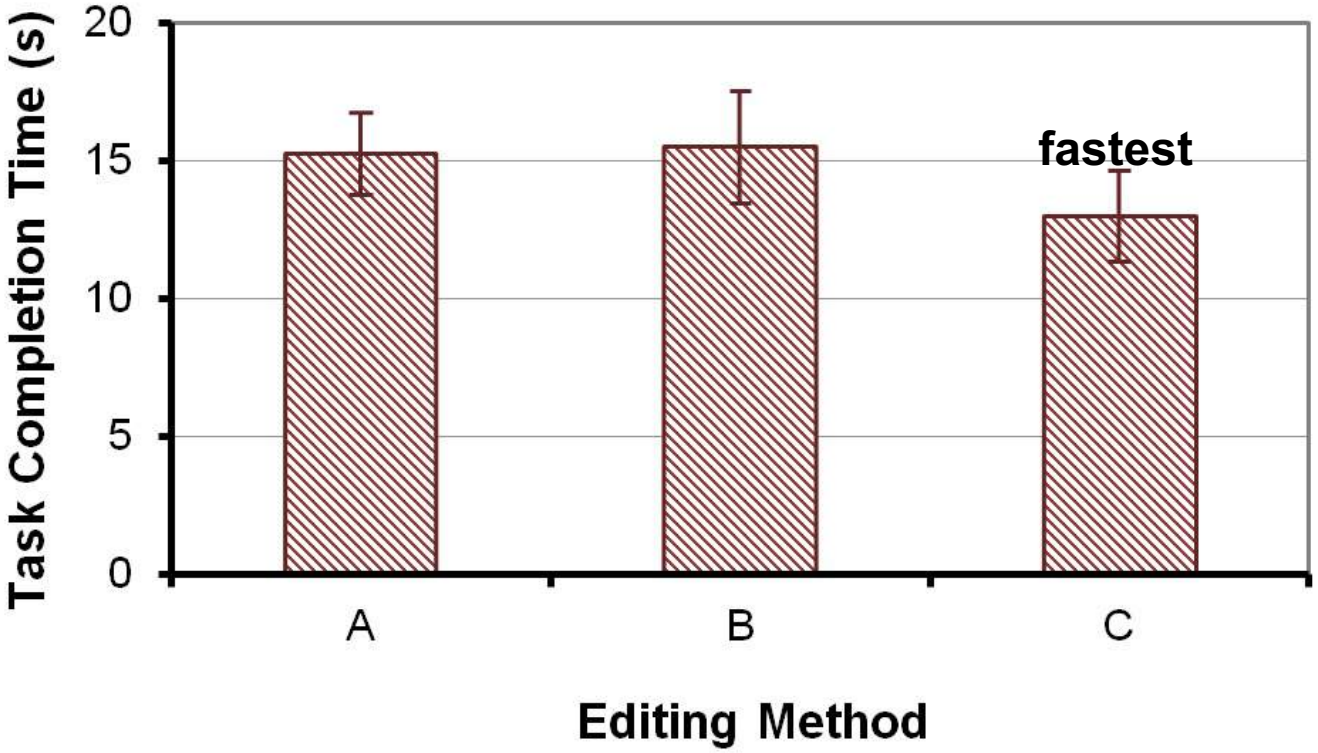
Results - Counterbalancing

	Participant	Test Condition			Group	*Mean*	*SD*
		A	B	C			
ABC	1	12.98	16.91	12.19	1	14.7	1.84
ABC	2	14.84	16.03	14.01			
ABC	3	16.74	15.15	15.19			
ABC	4	16.59	14.43	11.12			
BCA	5	18.37	13.16	10.72	2	14.6	2.46
BCA	6	15.17	13.09	12.83			
BCA	7	14.68	17.66	15.26			
BCA	8	16.01	17.04	11.14			
CAB	9	14.83	12.89	14.37	3	14.4	1.88
CAB	10	14.37	13.98	12.91			
CAB	11	14.40	19.12	11.59			
CAB	12	13.70	16.17	14.31			
	Mean	15.2	15.5	13.0			
	SD	1.48	2.01	1.63			

Group effect is small

∴ Counterbalancing worked!

Results - Counterbalancing

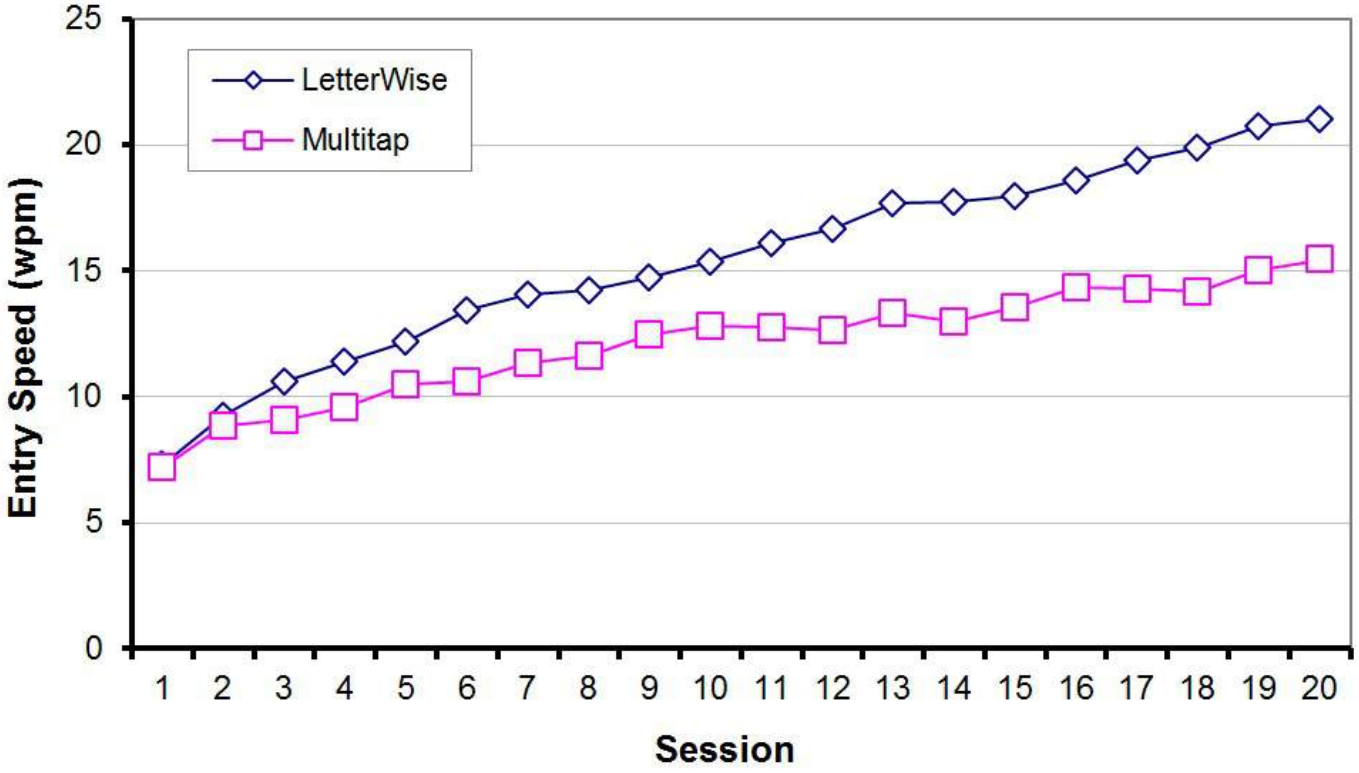


Longitudinal Studies

- Sometimes instead of "balancing out" learning effects, the research seeks to promote and investigate learning
- If so, a **longitudinal study** is conducted
- Practice: the independent variable (IV)
  - Participants are **practiced** over a prolonged period of time
  - Practice units: blocks, sessions, hours, days, etc.

[Longitudinal Study of Parents and Children]

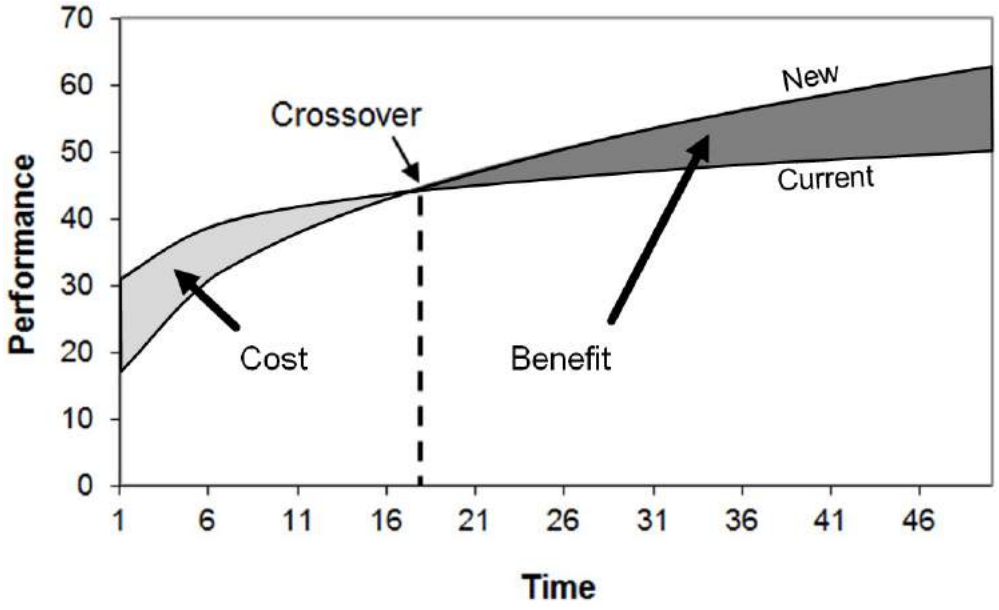
Longitudinal Study – Results[1]



[1] MacKenzie, I. S., Kober, H., Smith, D., Jones, T., & Skepner, E. (2001). LetterWise: Prefix-based disambiguation for mobile text entry. *Proceedings of the ACM Symposium on User Interface Software and Technology - UIST 2001*, 111-120, New York: ACM.

Cost-Benefit Trade-offs

- Sometimes, the benefit in the new technique is insufficient to overcome the cost in learning it



[Example: Smartphone Evolution]

[Example: Early Touch Screen Technology]

Review

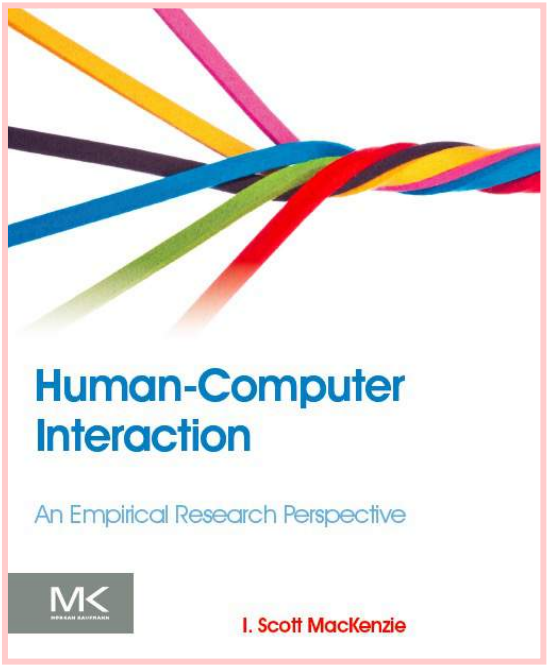




Francis Bacon

[Scientific Method and Experiment Design]

Thank You


Human-Computer Interaction
An Empirical Research Perspective
I. Scott MacKenzie